

## Scalability & Future Plan

|                      |                                       |
|----------------------|---------------------------------------|
| <b>Date</b>          | 04 July 2026                          |
| <b>Team ID</b>       |                                       |
| <b>Project Name</b>  | A Comprehensive Measure of Well-Being |
| <b>Maximum Marks</b> | 1 Mark                                |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                      | Impact                                     | Priority to Address<br>(High / Medium / Low) |
|------|-------------------------------------------------|--------------------------------------------|----------------------------------------------|
| 1    | Supports a limited number of simultaneous users | Performance may decrease with high traffic | High                                         |
| 2    | Uses a limited dataset for model training       | Prediction accuracy can be improved        | High                                         |
| 3    | Not deployed on a cloud platform                | Limited accessibility                      | Medium                                       |

### Scalability Plan:

| S.No | Scalability Aspect | Current State          | Proposed Upgrade / Solution                               |
|------|--------------------|------------------------|-----------------------------------------------------------|
| 1    | User Load          | Supports local users   | Deploy on a cloud platform with multi-user support        |
| 2    | Data Storage       | Local dataset          | Use a cloud database such as MySQL or PostgreSQL          |
| 3    | Performance        | Basic ML prediction    | Optimize the model and use caching for faster predictions |
| 4    | Security           | Basic input validation | Add user authentication and secure HTTPS communication    |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|-------|-------------------------------|-----------------|-----------------|
|-------|-------------------------------|-----------------|-----------------|

|         |                                                    |            |                                            |
|---------|----------------------------------------------------|------------|--------------------------------------------|
| Phase 1 | Improve prediction accuracy using a larger dataset | 1–2 Months | More accurate HDI predictions              |
| Phase 2 | Deploy the application on a cloud platform         | 2–3 Months | Access from anywhere                       |
| Phase 3 | Add user login and dashboard                       | 3–4 Months | Better user experience and security        |
| Phase 4 | Integrate data visualization and analytics         | 4–6 Months | Better insights through interactive charts |